Person-Name Extraction With Support Vector Machines:

Applying the CoNLL NER Corpus to the BNC Baby Corpus

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Named entity recognition (NER) is the task of finding spans of text that name real-world things and labeling each span with a type such as person, location, or organization (Nadeau & Sekine, 2007). This report describes how a support vector machine (SVM) trained on *Conll_NER_corpus.txt* can identify person names in the first 10,000 sentences of the BNC Baby corpus, and reports results from a working implementation.

Named Entities and the Training Data

Conll_NER_corpus.txt is derived from the Annotated Corpus for Named Entity Recognition on Kaggle (Walia, n.d.) and follows the CoNLL column format (Tjong Kim Sang & De Meulder, 2003): one token per line, a tab, and the token's tag, with a blank line between sentences. The file holds 47,959 human-tagged sentences and 1,048,575 tokens. Tags use the BIO scheme (Ramshaw & Marcus, 1995): *B-* marks the first word of an entity, *I-* marks a word inside it, and *O* marks a word outside any entity. Person names are tagged *B-per* (16,990 tokens) and *I-per* (17,251 tokens); in this corpus a title such as *President* or *Mr.* is part of the name, so "President Mahmoud Ahmadinejad" is tagged B-per I-per I-per. All other tags (geo, org, gpe, tim, art, eve, nat) are mapped to O, leaving three classes: B-PER, I-PER, and O.

Why an SVM

An SVM finds the hyperplane that separates two classes with the largest margin (Cortes & Vapnik, 1995). Linear SVMs handle the very large, sparse feature vectors produced by text well and were a strong NER baseline before neural models (Isozaki & Kazawa, 2002). With three tags, one binary SVM is trained per tag (one-versus-rest) and each token receives the tag with the highest score. The implementation uses the LIBLINEAR solver (Fan et al., 2008)

through scikit-learn's *LinearSVC* (Pedregosa et al., 2011), and NLTK's *BNCCorpusReader* to read the BNC XML files (Bird et al., 2009).

Procedure

The procedure has two phases, shown in Figures 1 and 2. In training (Figure 1), every token becomes a feature dictionary describing the word itself (lowercased form, first and last three letters, capitalization, all-caps, digits, a trailing period as in *Mr.*), its shape (e.g., *McCain* becomes *XxXx*), and the same information for the two words on each side, since context such as *said* or a preceding title signals a name. A vectorizer builds a sparse matrix, 10% of sentences are held out to measure precision and recall, and the final model is retrained on all sentences with balanced class weights, because O covers about 97% of tokens.

In extraction (Figure 2), the first 10,000 sentences (<*s*> units) of BNC Baby are read in file order, given the same features, and tagged; each B-PER and its following I-PER tokens form a candidate name. A regular expression keeps only spans that look like names (capitalized words, initials, hyphens, apostrophes, and inner particles such as *van* or *de*). Names are written with their sentence numbers, followed by a frequency list.

Pseudocode

```
PROCEDURE TrainPersonTagger(conllFile)
    sentences ← ReadSentences(conllFile)          // split at blank lines
    FOR each (word, tag) in sentences:
        tag ← 'B-PER' if tag = 'B-per', 'I-PER' if tag = 'I-per', else 'O'
    X, y ← [], []
    FOR each sentence s, FOR i ← 0 TO |s|-1:
        X.append(Features(s, i));  y.append(tag of s[i])
    vectorizer ← DictVectorizer().fit(X)
    (Xtr, ytr), (Xte, yte) ← Split(X, y, 90% / 10%)
    svm ← LinearSVC(C = 0.1, class_weight = balanced).fit(Xtr, ytr)
    Report precision, recall, F1 of svm on (Xte, yte)
    RETURN LinearSVC(...).fit(X, y), vectorizer  // retrain on all data

FUNCTION Features(s, i)
    f ← {lower(s[i]), shape(s[i]), prefix3, suffix3, isTitle, isUpper,
         isDigit, endsWithPeriod, isFirstWord}
    FOR k IN {-2, -1, +1, +2}:
        f ← f U {lower(s[i+k]), shape(s[i+k]), isTitle(s[i+k])} or <pad>
    RETURN f
```

```
PROCEDURE ExtractPersons(svm, vectorizer, bncDir, N = 10000)
    sents ← first N sentences of BNCCorpusReader(bncDir).sents()
    results ← []
    FOR k, s IN enumerate(sents):
        tags ← svm.predict(vectorizer.transform(Features(s, i) for each i))
        FOR span IN JoinSpans(s, tags):     // B-PER followed by I-PER*
            IF Matches(NAME_REGEX, span): results.append((k, span))
    Write results and name frequencies to person_names.txt
```

Figure 1
Training the SVM Person-Name Tagger on the CoNLL Corpus

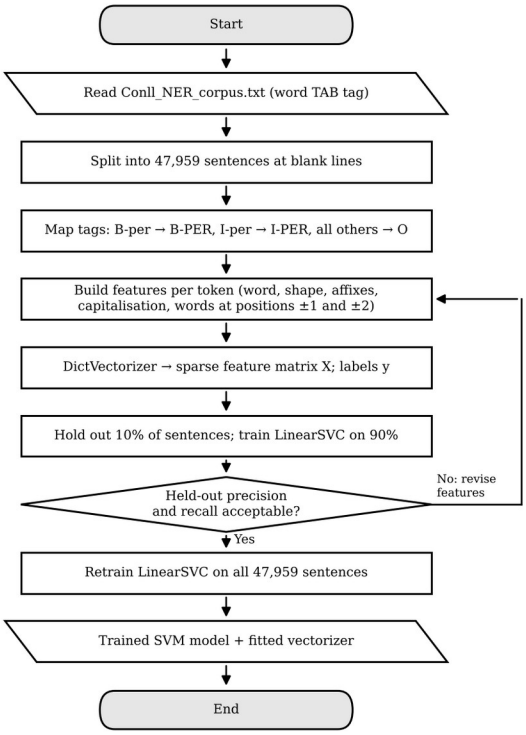
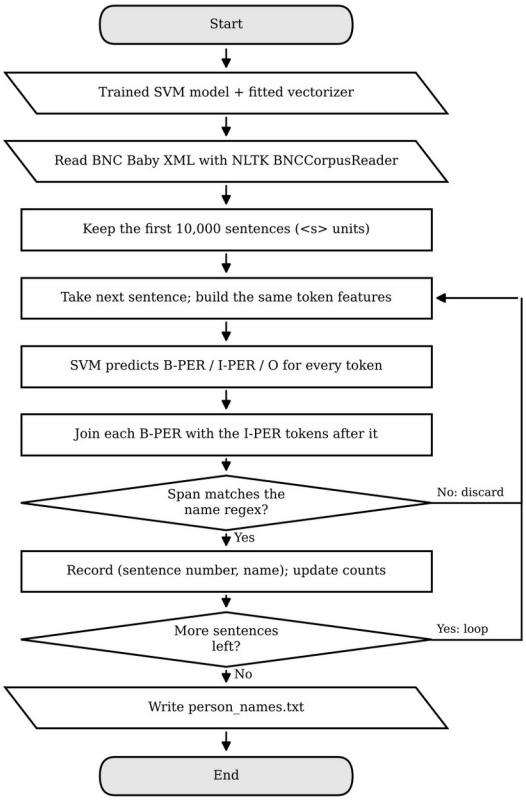


Figure 2
Extracting Person Names From the First 10,000 BNC Baby Sentences



Results

On the 10% held-out split, token-level scores for B-PER were precision .766, recall .876, and *F*1 .817; for I-PER they were .791, .926, and .853. Applied to the first 10,000 BNC Baby sentences (eight academic texts), the program extracted 1,213 person-name mentions covering 686 distinct names. The most frequent were Levin (20 mentions), Kahlo (16), Debord (16), Wilson (13), and Lord Scarman (13), alongside false positives such as *Second* (23), *GIS* (12), and *Monte Carlo* (12). As the lab allows, these errors were not corrected.

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